Kinetic Inductance in Superconductors

1 Introduction

In superconductors, electrical resistance vanishes, yet charge carriers (Cooper pairs) possess inertia. When a time-dependent current flows, energy is required to accelerate these carriers. This leads to an additional inductive effect known as **kinetic inductance**, distinct from geometric inductance.

2 Basic Derivation

Consider a superconducting wire of length l and cross-sectional area A, with Cooper pair density n_s, charge $q = 2e$, and effective mass m^*.

The current density is:

$$J = n_s q v_s \tag{1}$$

Total current:

$$I = n_s q A v_s \tag{2}$$

The kinetic energy per unit volume is:

$$\mathcal{E} = \frac{1}{2} n_s m^* v_s^2 \tag{3}$$

Total kinetic energy:

$$E_k = \frac{1}{2} n_s m^* A l \, v_s^2 \tag{4}$$

Substituting $v_s = \frac{I}{n_s q A}$:

$$E_k = \frac{1}{2} \frac{m^* l}{n_s q^2 A} I^2 \tag{5}$$

Comparing with inductive energy:

$$E = \frac{1}{2} L_k I^2 \tag{6}$$

we obtain the kinetic inductance:

$$\boxed{L_k = \frac{m^* l}{n_s q^2 A}} \tag{7}$$

3 Relation to London Penetration Depth

The London penetration depth is defined as:

$$\lambda_L^2 = \frac{m^*}{\mu_0 n_s q^2} \tag{8}$$

Thus:

$$\boxed{L_k = \mu_0 \lambda_L^2 \frac{l}{A}} \tag{9}$$

4 Thin Film Limit

For thin films with thickness $d \ll \lambda_L$, the sheet kinetic inductance is:

$$L_k^\square = \frac{\mu_0 \lambda_L^2}{d} \tag{10}$$

5 Frequency Dependence

The complex conductivity is:

$$\sigma(\omega) = \sigma_1(\omega) - i\sigma_2(\omega) \tag{11}$$

The kinetic inductance is related to the imaginary part:

$$L_k = \frac{1}{\omega \sigma_2} \tag{12}$$

6 Nonlinear Kinetic Inductance

At high currents:

$$L_k(I) \approx L_k(0) \left(1 + \alpha \frac{I^2}{I_c^2}\right) \tag{13}$$

where I_c is the critical current.

7 Temperature Dependence

The superfluid density decreases with temperature:

$$n_s(T) \downarrow \quad \Rightarrow \quad L_k(T) \uparrow \tag{14}$$

Near the critical temperature T_c, the kinetic inductance diverges.

8 Applications

- Kinetic Inductance Detectors (KIDs)
- Superinductors in quantum circuits
- Parametric amplifiers
- Superconducting qubits
- High-impedance transmission lines

9 Conclusion

Kinetic inductance arises from the inertia of Cooper pairs in a superconductor. It becomes significant in systems with low superfluid density or reduced dimensionality, and plays a central role in modern superconducting quantum technologies.